

The power of music

by Brid O'Riordan



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Here is an essay from one of our followers Brid O'Riordan, on the power of music on regulating and improving mood, highlighting the science behind the thinking

Music therapy is fast becoming a very well recognised treatment for those with mental health issues.

In this essay I will be discussing the power of music and its beneficial therapeutic effects on human emotions and mood. Music is a universal language that can be understood by all humans it – 'dynamically engages the emotional and spiritual part of being human' It offers comfort and therapy to those suffering conditions such as brain damage, dementia, Alzheimer's Disease, autism and depression.

Music can have a profound influence on mood, potentially elevating the positive aspects of mood, such as vigour, excitement and happiness, and reducing depression, tension, fatigue, anger and confusion.

I will relate some personal experiences where music provided a medium for communication amongst my own family members who had limited brain function, providing insight into the parts of the brain associated with emotions and mood and how music helps improve them. I will discuss music therapies and the ways they can be effective in helping people in both a physiological and psychological manner.

Margaret, the eldest member of my family unit, was severely handicapped, blind, and yet a very happy person. Listening to Radio Luxembourg and RTE's Gerry Ryan were her two greatest pleasures in life. Radio Luxembourg helped her to sleep at night and Gerry Ryan kept her happy during the morning. Music was her first love. She knew all the latest pop songs and we all sang along with her. Even though she was challenged both mentally and physically she

used music as a form of communication with everyone and used it well! She was happy when she sang and music made her happy again when she felt sad.

My mother Bridie, suffered a stroke, which induced aphasia, which is a difficulty understanding words. She could however vocalise two words – ‘yes’ and ‘no’, but they were not always appropriately placed in her responses. She also had paralysis of the right side of her body which meant the left side of her brain was damaged. A lengthy recovery period in hospital followed. The first morning following her arrival home from hospital was most memorable. Her frustrating inability to express language since her stroke emphasised the shock and delight expressed both on her face and ours when she suddenly started to sing along with ‘The Rose of Tralee’ as it played on the radio.

Thankfully amusia, which is the loss in understanding music, did not affect either Margaret or my mother, as music played such a fundamental part in their lives. It provided relief for both of them in times of frustration and stress. It also provided a medium in which we all could sing together offering a method of communication and harmony. Music helped alleviate the symptoms of depression that became very prominent in my mother following her stroke.

Personal interest arose in the therapeutic benefits of music after suffering a period of disharmony within myself, caused from a build-up of stress due to work, self-employment, and family commitments and so on, resulting in a lack of self-care. As a result I became overwhelmed. Music was a help at the time and offered an escape from the complicated workings of my mind. Playing and singing raised my spirits and at least gave relief from the sterility of not caring about anything or anybody. Depression is cruel and unforgiving; it consumes and negates all thoughts to the point of exhaustion. A realisation period has to be endured in order to re-evaluate and re-formulate one’s circumstances. The paradox is music helped me become more in touch with my feelings while also offering relief from them. This was the first step to healing, which I was later to realise through literature and therapy. But how does music help people heal?

The nervous system is the body’s decision and communication centre. The central nervous system is made up of the brain and spinal cord and the peripheral nervous system comprises the other associated nerves.

The brain is the control centre for all the body’s functions, such as walking, talking, swallowing, breathing, taste, smell, heart rate and so on. It also controls all our thinking functions, our emotions, how we behave and all our intellectual (cognitive) activities, such as how we attend to things, how we perceive and understand our world and its physical surroundings, how we learn and remember and so on.

The brain is divided into left and right hemispheres. The left brain is responsible for words, logic, numbers and analysis. It controls the right side of the body. The right hemisphere is the creative side and is responsible for rhythm, colour, imagination, awareness and dimension. It controls the left side of the body. Each of these hemispheres is responsible for the multitude of specific systems required for functioning. Practically every part of the brain is involved while listening, playing, reading and creating music. It affects both sides of the brain at once thus allowing the brain to concentrate more easily. Listening involves memory centres of the brain while playing music requires motor control and co-ordination of hand movements in the case of musicians. Reading music involves the visual cortex. The part of the brain most associated with emotion is called the amygdala; it is responsible for decision – making, negative emotions and the appropriate reactions to situations. It is also responsible for which memories are stored and in which part of the brain. I find this interesting as music can awaken memories of one’s specific location in time and place when played. The brain contains hormones ‘A hormone is a chemical the body produces that controls and regulates certain activities in cells or organs’. Some of these related to emotions include:

Cortisol regulates levels of stress.

Severe and prolonged stress can impair the ability of the brain to function normally for a period of time, but the brain is also capable of remarkable recovery.

Music seems to affect 'particularly the "stress" hormone cortisol' in that it can reduce its levels.

Serotonin is associated with mood, appetite, sleep, and depression. It principally tends to have an inhibitory effect on these functions. Natural daylight and regular exercise help increase serotonin levels. Stress reduces these levels. Music increases the levels.

Dopamine is associated with emotional highs and lows, movement, sleep, memory and pleasurable reward. Reduced levels are thought to be associated with Parkinson's disease. It causes the brain to actively seek out pleasurable addictions like food, drugs and other stimulants to cause its release in the brain. When someone experiences a 'feel-good' feeling in anticipation of listening to their favorite music it causes the release of dopamine. 'It is amazing that we can release dopamine in anticipation of something abstract, complex and not concrete,' Salimpoor said. 'This is the first study to show that dopamine can be released in response to an aesthetic stimulus.'

Adrenalin determines 'fight or flight'. It allows the body to react quickly to a stressful situation while also reducing pain threshold levels. Prolonged high levels of adrenalin in the blood can lead to insomnia and nervous behaviors which can be an indication of chronic stress. Exercise can 'burn-off' some of the extra levels. Music offers relaxation.

What is Music? Music is a universal language understood by all people. Music involves practically every part of the brain. It evokes emotional responses from everyone irrespective of one's physical or mental capacities. It channels pathways for communication not involving linguistic skills. Children who learn music are said to have better literacy, numeracy, problem solving, disciplinary and social skills. It allows for the expression of emotion that some individuals may find difficult to express through language. It has therapeutic benefits that can help with healing. Anthony Storr states 'some philosophers have thought that music enabled a listener to escape the pains of existence by temporarily entering a realm of peace' And Oliver Sacks in his book Musicophilia expands this belief when stating 'music can also evoke worlds very different from the personal, remembered worlds of events, peoples, places we have known' Therefore it is very evident that music has a power that is both unique and universal.

Three methods applied in providing music as therapy and healing? These include music therapy, sound therapy and Mozart therapy or 'the Mozart Effect':

I conducted an interview with Dara O' Brien who works as a music therapist with autistic children. I asked him to define music therapy. 'Generally it means, in the western sense, quite a clinical approach to using music as an intervention for people's wellbeing' He believes that music does harmonise mind, body and spirit, and it can somehow take people into a different space. Music therapy is becoming increasingly recognised in the treatment of mental and neurological disorders. It has shown effectiveness in treating dementia, Alzheimer's, stroke, aphasia, and depression among many others. Music therapy can help in the recovery of motor skills. In some cases it has become a 'reminder' memory of daily routine for many people suffering with memory loss. Activities, for example, getting out of bed, putting on clothes, eating meals can be carried out by people while listening to certain melodies which have been learned to associate and enable procedure of these daily routines.

Sound therapy helps connect different parts of the brain. It can enhance sleep, work, concentration and communication. It helps clear thinking and makes people calmer and happier. "Sound healing is the practice of using audio tones and vibrational frequencies to repair damaged tissue and cells within the body." Tibetan singing bowls were used in ancient India going back 4000 years. They are effective because they produce high frequency sounds

which stimulate middle ear activity, and once the ear is open people can experience deep meditation, thus ensuing healing at a deeper level. Sound Therapy and meditative practices are thought to stimulate feelings of happiness and peace located in the left brain. The stimulation of the left brain through these methods is thought to have a beneficial impact on depression and anxiety. During this high frequency therapy it is thought that the hormones related to improved mood are stimulated.

Some people say that the music of Mozart has perfection in its compositional integrity and is universally beautiful. It can have an amazingly powerful psychological effect on people. Mozart's compositions are highly repetitive and therefore may affect the organizational center of the brain. Some researchers believe that 'concentration, reasoning and even the ability of the human body to heal itself' can happen from listening to Mozart's music. Many researchers believe that Mozart's music has a significant effect on the neurological activities of the human brain, and some believe the 'Mozart Effect' 'can increase their 'spatial intelligence' simply by listening to the composer's music'. Brenda Fricker, a renowned Irish actress recently reported getting relief from her depressive episodes by attending a doctor who listens to the patient and then selects certain pieces from Mozart's repertoire. But can it be argued that listening to Mozart music alone cannot help everyone, because musical taste is very individual and varied that the music has to be enjoyed by the person for them to receive therapy from the music.

Three areas that receive symptom relief and release through music are; autism, brain degenerative conditions like dementia and Alzheimer's Disease as well as mental illnesses such as depression.

An autistic child has impaired social and communicative characteristics and can repeat certain behaviours continuously. Dara O' Brien in his work finds the 'act of listening' can be very therapeutic because it centres and relaxes the children. He finds that music offers the children a gateway to express their frustrations and states that it can give them a clear pathway in understanding their feelings of sadness when 'sad' music is played and similarly feelings of joy and happiness when 'cheerful' music is played. I experienced the benefit of music while sitting with an autistic child who was quite agitated and uneasy. There were lots of people in the room and I was aware that this might be some of the cause for this behaviour. I began to sing lullaby's into her ear and after a little while she began to hum along and her agitation reduced significantly. She cannot speak cognitively but walks around most of the time now with headphones playing tunes from 'One Direction', thus keeping her calm!

'Depression is any attempt by the mind and consequently the body to repress the natural state and spirit of a person'. This definition was given by Eamon Keane, a counsellor and musician, during an interview I conducted with him. He further states that it is also, 'loss and unresolved grief – and peoples struggle to be who they really are and who they are meant to be'. Depression is directly related to levels of serotonin and dopamine. In people that have an imbalance in hormone levels, medicine is necessary. However, I believe that drugs are being prescribed too easily today in the treatment of reactive depression and anxiety. Prescription drugs only suppress feelings with the result that patients delay their mental care. Everybody feels low or sad at some stage in their lives, and no more than 'building-up' antibodies in one's body in order to fight off a flu virus, can methods of fighting off feelings of 'blues' be learned. Oliver Sacks tells of one young man who had bipolar disorder and found that he was able to regulate his mood swing by playing the piano, and finds he uses it as some people use medications to stabilize their moods. 'Tchaikovsky and Berlioz, when depressed, found music to be a life-saver' Eamon Keane also says that emotional trauma causes injury to the brain. It is thought that with the advancement in brain scanning (fmri) areas of damage can be identified and repaired with treatment. Can one infer thus that music, because we already know it does provide brain stimulation while learning, playing and reading, can therefore act as a stimulant for these new neural pathways therefore ensuring its significance in the healing process?

Oliver Sacks is a neuroscientist who works predominantly with patients suffering with varying forms of brain disorders including dementia and Alzheimer's. In his book 'Musicophilia' he tells of one man, Clive Wearing, an English musician and musicologist who was struck by a brain infection, herpes encephalitis that affected parts of his brain concerned with memory. Clive was left with a memory span of only a few seconds. However, even in this amnesiac state he retained his ability to read and play music with equal competence prior to his illness. 'The embedding of words, skills, or sequences in melody and meter is uniquely human'. Music is a language known to all; it forms a springboard for communication which in turn complements healing. He says that all patients suffering with dementia – without exception – retain their musical powers and tastes even when most other mental powers have been severely compromised. They can recognise music and respond to it emotionally even when little else can get through.

He also tells of an elderly lady who presented quite bothered and agitated with the effects of Alzheimer's. It was found that by placing a MP3 player over her ears playing her favourite tunes kept her in a calm state. Sack's says 'That even in a diffuse cortical disease like Alzheimer's, music can still be perceived,' This, for me, provides fascinating facts where patients with such degenerative brain conditions can still quantify and appreciate music. A comparison can be made here between the autistic girl and this lady with Alzheimer's. Both have quite different neurological disability but each enjoy their favourite music through headphones helping to keep them calm and peaceful.

In conclusion music can be medicine for soul and mind. Music is so innately part of humans and is practically a guaranteed way in raising people's spirits irrespective of their intellectual cognitive abilities. Can one infer that the therapy offered by music when mixed with exercise and counselling on a regular basis could offer relief to individuals experiencing feelings of depression and anxiety? It worked for me. Music offers therapy through reprieve from thoughts and emotions either by listening or playing. It also can alter hormone levels concerned with emotion. Perhaps it may be possible that someday music will hold significance amongst the medical world for the therapeutic attributes it has to offer.

We can expect that a process of research, reflection and review, modification and innovation, and further research will continue to enrich our collective understanding of the untapped potential that music holds for health and healing and will thus give rise to new models of this potential and the new ways to more fully embrace it.

By introducing correct methods of therapy, repair and change can occur. Perhaps music may become a recognised medicine for the soul and mind.

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Article by Brid O'Riordan

A Lust For Life reader and follower. A Lust For Life is a movement for well-being. We believe in the power of creating and sharing information that can help us all navigate this sometimes difficult but often wonderful world. We know that in order to live well we need to look at life holistically, at all aspects of what it means to be human.